Thm. $U \otimes_R -$ the functor $U\otimes_R - : R\text{-Mod} \to S\text{-Mod}$ is right exact

Pf Let $N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$ be exact g is epi, $\mathrm{im} f = \ker g$

Aim: prove $U\otimes_R N' \xrightarrow{1\otimes f} U\otimes_R N \xrightarrow{1\otimes g} U\otimes_R N'' \to 0$ exact $\Rightarrow$ show

$1\otimes g$ is epi. $\mathrm{im}(1\otimes f) = \ker(1\otimes g)$

(1) $1\otimes g$ is epi.

(2) Prove: $\mathrm{im}(1\otimes f) = \ker(1\otimes g)$

- $(1\otimes g)\circ(1\otimes f) = 1\otimes gf$, since $gf=0$, we have $1\otimes gf = 0$

hence $\mathrm{im}(1\otimes f) \subseteq \ker(1\otimes g)$

- Show the inverse inclusion

φ $U\otimes_R N / \mathrm{im}(1\otimes f) \longrightarrow U\otimes N''$

$\overline{x\otimes y} \longmapsto x\otimes g(y)$

$\mathrm{im}(1\otimes f) = \ker(1\otimes g)$ if and only is this is an isomorphism.

Let $x\in U$, $y''\in N''$ choose $g(y) = y''$ for some $y\in N$

$(1\otimes g)(x\otimes y) = x\otimes y''$

claim: $\overline{x\otimes y} \in U\otimes_R N/\mathrm{im}(1\otimes f)$ is independent of the choice of y

Let $y_1, y_2 \in N$ s.t. $g(y_1) = g(y_2) = y''$ $\quad y_1 - y_2 \in \ker g = \mathrm{im} f$

$\Rightarrow y' \in N'$ s.t. $y_1 - y_2 = f(y')$

$x\otimes y_1 - x\otimes y_2 = x\otimes(y_1-y_2) = x\otimes f(y') \in \mathrm{im}(1\otimes f)$

hence $\overline{x\otimes y_1} = \overline{x\otimes y_2}$ in $U\otimes_R N/\mathrm{im}(1\otimes f)$

Therefore $U\times N'' \longrightarrow U\otimes_R N/\mathrm{im}(1\otimes f)$ blanced map.

$(x, y'') \longmapsto \overline{x\otimes y}$

So we have a homo. $\theta' : U \otimes_R N'' \longrightarrow U \otimes_R N / \operatorname{Im}(1 \otimes f)$

$$x \otimes y'' \longmapsto \overline{x \otimes y}$$

check that $\theta \theta' = 1_{U \otimes_R N''}$, $\theta' \theta = 1_{U \otimes N / \operatorname{Im}(1 \otimes f)}$

<u>Thm</u> ${}_R U_S$ The functor $- \otimes_R {}_R U_S : \text{Mod-}R \longrightarrow \text{Mod-}S$

is right exact.

<u>Def.</u> ${}_R M$ is called projective if $\operatorname{Hom}_R(M, -)$ is exact

is injective if $\operatorname{Hom}_R(-, M)$ is exact.

flat if $- \otimes_R M$ is exact.

<u>Thm.</u> The following statements about left R-module P are equivalent:

(1). whenever there is given a diagram

$$\begin{tikzcd} & P \arrow[dl, dashed, "\bar{\gamma}"'] \arrow[d, "\gamma"] & \\ M \arrow[r, "g"'] & N \arrow[r] & 0 \end{tikzcd}$$

where g is an epimorphism, there is an R-homomorphism

$\bar{\gamma} : P \to M$ s.t. $\gamma = g \bar{\gamma}$

(2). for each epi $f : {}_R M \to {}_R N$, the map

$\operatorname{Hom}_R(P, f) : \operatorname{Hom}_R(P, M) \to \operatorname{Hom}_R(P, N)$ is an epi

(3) For each bimodule structure ${}_R P_S$ the functor

$\operatorname{Hom}_R(P, -)$ $R\text{-Mod} \to S\text{-Mod}$ is exact.

(4). For every exact sequence $M' \xrightarrow{f} M \xrightarrow{g} M''$ in R-mod

the sequence $\operatorname{Hom}_R(P, M') \xrightarrow{\operatorname{Hom}_R(P, f)} \operatorname{Hom}_R(P, M) \xrightarrow{\operatorname{Hom}_R(P, g)} \operatorname{Hom}_R(P, M'')$

is exact.

pf: (1) (2)
(2) (3) clear

(3) (4) exercise.

A left R-module P is projective if it satisfies one of the above statement.

Free module are projective.

<u>Thm</u> The following statements about a left R-mod P are equiv

(1). P is projective

(2). Every epimorphism ${}_RM \xrightarrow{f} {}_RP \to 0$ is split.

that is, there is $g: P \to M$ a homo. s.t. $fg = 1_P$

(3). P is isomorphic to a direct summand of a free module

<u>pf</u>: (3) $\Rightarrow$ (1) ✓

(2) $\Rightarrow$ (3). Since every module is a homo image of some free module.

$$\underset{\text{free}}{F} \xrightarrow{f} P \to 0 \qquad f \text{ epi}$$

$$\exists g, \quad fg = 1_P. \quad \Rightarrow F = \ker f \oplus g(P) \qquad g(P) \cong P.$$

(1) $\Rightarrow$ (2)

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "1_P"] & \\ M \arrow[r, "f"] & P \arrow[r] & 0 \end{tikzcd} \qquad 1_P = fg$$

Corollary: A left R-module P is finitely generated and projective if and only if for some R-module P' and integer $n > 0$ there is an R-Isomorphism $P \oplus P' \cong R^n$.

Thm. (Schanuel Lemma) Let $0 \to K \to P \xrightarrow{\lambda} V \to 0$ and $0 \to K' \to P' \xrightarrow{\lambda'} V \to 0$ be short exact sequences, with P, P' projective. Then $K' \oplus P \simeq K \oplus P'$

prf: Let $Y = \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\}$

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
 & & & K \arrow[d] & \\
 & & Y \arrow[r, "u"] \arrow[d, "u'"'] & P \arrow[d, "\lambda"] & \\
0 \arrow[r] & K' \arrow[r] & P' \arrow[r, "\lambda'"] & V \arrow[r] \arrow[d] & 0 \\
 & & & 0 &
\end{tikzcd}$$

we call Y the pull back of (λ, λ')

$u : Y \to P \quad (p, p') \mapsto p \qquad u' : Y \to P' \quad (p, p') \mapsto p'$

$\ker u = \{(0, p') \mid \lambda'(p') = 0\} \simeq K'$

$\ker u' = \{(p, 0) \mid \lambda(p) = 0\} \simeq K$

So we have two short exact sequences.

$0 \to K' \to Y \xrightarrow{u} P \to 0$

$0 \to K \to Y \xrightarrow{u'} P' \to 0$

both split since P, P' are projective

In particular, $K' \oplus P \simeq K \oplus P'$

Lemma: ${}_R M$: left module. Then $M \simeq \mathrm{Hom}_R({}_R R_R, M)$ as left R-modules.

proof: $\mathrm{Hom}_R(R, M) \longrightarrow M$

$f \longmapsto f(1)$

$(r \mapsto rm) \longleftarrow m$

Corollary: (For any ring with 1) $\mathrm{End}_R(_RR) \simeq R^{op}$

pf: $\mathrm{End}_R(R) \longrightarrow R^{op}$

$f \longmapsto f(1)$

op here since $f_1 f_2(1) = f_1(f_2(1) \cdot 1) = f_2(1) f_1(1)$

Take $\mathrm{Hom}_R(M, R) = M^*$.

Let R be a commutative ring with 1.

For R-modules X, Y and M, we have an R-homomorphism.

$$\mathrm{Hom}_R(X, M) \otimes \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y)$$

$$\varphi \otimes \psi \longmapsto \psi\varphi$$

Lemma: Then image of the abover morphism is comprised of those morphism from X to Y which factor through som finite power of M

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \\ & M^n \arrow[ur] & \end{tikzcd}$$

Pf. An element of the image is $\sum_{i \in I} \psi_i \varphi_i$ $|I| < \infty$

$\varphi_i \in \mathrm{Hom}_R(X, M)$ $\psi_i \in \mathrm{Hom}_R(M, Y)$

correspond to a morphism

$$\varphi : X \to M^I$$

a morphism

$$\psi : M^I \to Y$$

Take $M=R$ we get a morphism

$$T_{X,Y}: X^* \otimes_R Y \longrightarrow Hom_R(X,Y)$$

Consider $im(T_{X,Y})$ $f \in im(T_{X,Y})$ if f factor through a fin. rand free R-module

$\Rightarrow$ if f f.t a f.g. proj. module

Remove by $Hom_R^{pr}(X,Y)$ the image of $T_{X,Y}$, and its elements of $Hom_R^{pr}(X,Y)$ are called the projective morphism from X to Y

projective morphism = factor through a f.g. projective module

Lemma: For any R-module M, $Hom_R^{pr}(M,M)$ is a 2-side ideal of the ring $End_R(M) = Hom_R(M,M)$

Thm: Let P be an R-module TFAE

(1). P is a f.g. proj. module

(2). $Hom_R^{pr}(P,P) = End_R(P)$

(3). For any R-module X, the morphism

$T_{X,P}: X^* \otimes_R P \longrightarrow Hom_R(X,P)$ is an isomorphism.

(4). For any R-module X, the morphism

$T_{P,X}: P^* \otimes_R X \longrightarrow Hom_R(P,X)$ is an isomorphism.

(5) the morphism $T_{P,P}: P^* \otimes_R P \longrightarrow End_R(P)$ is an isomorphism.

(1) $\Rightarrow$ (2) clear

(3) $\Rightarrow$ (5), (4) $\Rightarrow$ (5) clear.

(2) $\Rightarrow$ (3) $Id_P = T_{P,P}\left(\sum_i \varphi_i \otimes m_i\right)$ $\varphi_i \in P^*$, $m_i \in P$

$\forall\, v \in P \qquad v = Id_P(v) = \sum_i \varphi_i(v) m_i$

Consider the morphism $\sigma: Hom_R(X, P) \longrightarrow X^* \otimes_R P$.

$$\alpha \longmapsto \sum_i \varphi_i \alpha \otimes m_i$$

check: σ is the inverse of $\rho_{X,P}$

- $Hom_R(X,P) \xrightarrow{\sigma} X^* \otimes_R P \xrightarrow{\rho_{X,P}} Hom_R(X,P)$

$$\alpha \longmapsto \sum_i \varphi_i \alpha \otimes m_i \longmapsto \left(x \longmapsto \sum_i \varphi_i(\alpha(x)) m_i \right.$$

- $X^* \otimes_R P \xrightarrow{\rho_{X,P}} Hom_R(X,P) \xrightarrow{\sigma} X^* \otimes_R P$

need to prove $\psi \otimes v = \sum_i \varphi_i \rho_{X,P}(\psi \otimes v) \otimes m_i$

$$\varphi_i(v)\psi = \varphi_i \rho_{X,P}(\psi \otimes v)$$

$$\psi \otimes v = \psi \otimes \sum_i \varphi_i(v) m_i = \sum \psi \otimes \varphi_i(v) m_i$$

$$= \sum_i \varphi_i(v)\psi \otimes m_i = \sum_i \varphi_i \rho_{X,P}(\psi \otimes v) \otimes m_i$$

(2) $\Rightarrow$ (4) similar.

(5) $\Rightarrow$ (1) In particular Id_P is a projective morphism

$$\begin{tikzcd} P \arrow[r, "Id_P"] \arrow[d, "\iota"'] & P & Id_P = \pi \circ \iota \\ R^n \arrow[ur, "\pi"'] & & \end{tikzcd}$$

$\Rightarrow P$ is a direct summand of R^n

$\Rightarrow P$ is a f.g. projective module.

R: ring with 1, not necessary commutative.

projective cover:

Def: An essential epimorphism is an epimorphism of modules $f: U \to V$ with property that no proper submodules of U

is mapped surjectively onto V by f

Equivalently whenever $g: W \to U$ is a morphism s.t

$f \cdot g$ is an epi the g is an epi

Def. A projective cover of a module U is an essential epimorphism

$P \twoheadrightarrow U$ where P is projective.

prop: sps $f: U \to V$, $g: V \to W$ are two R-homo.

If two of $f \cdot g$ gf are essential then so is the third

Pf. sps. $f \cdot g$ are essential epis, gf is also epi.

$U \xrightarrow{f} V \xrightarrow{g} W$ let U_0 be a proper submodule

of U. f essential $\Rightarrow$ $f(U_0) \neq V$

g essential $\Rightarrow$ $gf(U_0) \neq W$.

- sps f and gf are essential epis.

Since gf epi $\Rightarrow$ g epi

Let V_0 be a proper submodule of V

$f^{-1}(V_0) \neq U$ $g(V_0) = gf(f^{-1}(V_0))$ is a proper submod

of M

- sps that g and gf are essential epis

If f is not epi $f(U) \neq V$ $gf(U) \neq W$ since gf essential

hence. f is epi.

Let U_0 be a proper submodul of U

$gf(W_0) \neq W \qquad f(V_0) \neq V \qquad \square$

Def. M: R-module M is called Noetherian if for any chain of submodules $L_1 \subseteq L_2 \subseteq \cdots \subseteq \cdots$ there exists n s.t $L_{n+i} = L_n$ for all $i = 1, 2, \ldots$

Lemma. For a Module M. TAFE

(1). M is Notherian

(2) Every submodule of M is f.g

(3) Every non-empty set of submodules of M has a maximal element

Prop. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of R-modules. Then M is noetherian if and only if M' and M'' are both noetherian.

Def Artin

prop Artin.

Lem. An R-module M is Artinian iff every non-empty set of submodules of M contains a minimal element w.r.t inclusion.

If M is a Noetherian (or Artinian) module, we also say there M satisfies the maximal (or minimal) condition on submodules.

If . . . . . ACC / DCC . . . .

An R-module M is called a simple (or irreducible) module if it has exactly submodule : $0, M$